

Primary sources & further reading

Part 02 — 🧠 Deep Learning · 02-deep-learning/papers/PAPERS.md · 02-deep-learning/resources/RESOURCES.md

Papers — Deep Learning

`bash ../_tools/fetch_papers.sh 02-deep-learning` downloads the arXiv PDFs below into this folder.

The canon (read in this order)

#	Paper	Year	Why it matters	Link
1	ImageNet Classification with Deep CNNs (AlexNet) — Krizhevsky, Sutskever, Hinton	2012	The paper that started the deep learning era. GPUs + ReLU + dropout + data.	PDF
2	Deep Residual Learning (ResNet) — He et al.	2015	$F(x) + x$. Made 100+ layers trainable. Lives inside every transformer.	arXiv:1512.03385
3	Batch Normalization — Ioffe & Szegedy	2015	Normalisation as a training stabiliser; the "internal covariate shift" story was later disputed — read the critique too.	arXiv:1502.03167
4	Layer Normalization — Ba, Kiros, Hinton	2016	Batch-independent normalisation — what transformers actually use.	arXiv:1607.06450
5	Adam: A Method for Stochastic Optimization — Kingma & Ba	2014	The default optimizer.	arXiv:1412.6980
6	Decoupled Weight Decay Regularization (AdamW) — Loshchilov & Hutter	2017	The fix that trains every modern LLM.	arXiv:1711.05101
7	Dropout — Srivastava et al.	2014	Regularisation by random ablation.	PDF

Going deeper

Paper	Year	Takeaway	Link
Long Short-Term Memory — Hochreiter & Schmidhuber	1997	Gating lets gradients survive long sequences	PDF
Sequence to Sequence Learning with Neural Networks — Sutskever et al.	2014	Encoder–decoder; the shape of machine translation	arXiv:1409.3215
Neural Machine Translation by Jointly Learning to Align and Translate — Bahdanau et al.	2014	Attention is invented here , three years before the transformer	arXiv:1409.0473
Understanding the difficulty of training deep feedforward networks — Glorot & Bengio	2010	Xavier initialization	PDF
Delving Deep into Rectifiers (He init, PReLU) — He et al.	2015	The ReLU-appropriate init	arXiv:1502.01852
Auto-Encoding Variational Bayes (VAE) — Kingma & Welling	2013	Probabilistic latent spaces → Module 05	arXiv:1312.6114
Random Search for Hyper-Parameter Optimization — Bergstra & Bengio	2012	Why grid search wastes your budget	PDF
Mixed Precision Training — Micikevicius et al.	2017	bf16/fp16 training, loss scaling	arXiv:1710.03740
Bag of Tricks for Image Classification — He et al.	2018	The unglamorous tweaks that actually move accuracy	arXiv:1812.01187
A ConvNet for the 2020s (ConvNeXt) — Liu et al.	2022	CNNs modernised; a fair fight against transformers	arXiv:2201.03545
Deep Learning (Nature review) — LeCun, Bengio, Hinton	2015	The field's own summary of itself	DOI

Free books

Book	Link
Dive into Deep Learning (D2L) — interactive, PyTorch/JAX/TF	https://d2l.ai/
Deep Learning — Goodfellow, Bengio, Courville	https://www.deeplearningbook.org/
Neural Networks and Deep Learning — Michael Nielsen (best backprop chapter anywhere)	http://neuralnetworksanddeeplearning.com/
Understanding Deep Learning — Simon Prince (2023, free PDF + notebooks)	https://udlbook.github.io/udlbook/
The Little Book of Deep Learning — François Fleuret	https://fleuret.org/francois/lbdl.html
Probabilistic ML: Advanced Topics — Murphy	https://probml.github.io/pml-book/book2.html

Resources — Deep Learning

Courses

Course	Why	Link
Karpathy — Neural Networks: Zero to Hero	Builds autograd, then an LLM, from nothing. The best deep learning teaching on the internet.	https://karpathy.ai/zero-to-hero.html
fast.ai — Practical Deep Learning for Coders	Top-down; you train a good model in lesson 1	https://course.fast.ai/
DeepLearning.AI Deep Learning Specialization	Andrew Ng's structured treatment	https://www.coursera.org/specializations/deep-learning
Stanford CS231n — CNNs for Visual Recognition	The best notes on backprop and convnets	https://cs231n.github.io/
NYU Deep Learning (LeCun & Canziani)	Free lectures, strong theory	https://atcold.github.io/NYU-DLSP21/
MIT 6.S191 Intro to Deep Learning	One-week intensive, updated yearly	http://introtodeeplearning.com/
PyTorch official tutorials	The reference you will actually return to	https://pytorch.org/tutorials/

Repositories worth cloning

Repo	What's inside
https://github.com/karpathy/micrograd	A 100-line autograd engine. Read it in one sitting.
https://github.com/karpathy/nn-zero-to-hero	Notebooks for the course above
https://github.com/pytorch/examples	Canonical training scripts (MNIST, ImageNet, DCGAN, ...)
https://github.com/d2l-ai/d2l-en	Dive into Deep Learning, full source
https://github.com/rasbt/deeplearning-models	~60 clean reference implementations
https://github.com/lucidrains/vit-pytorch	Readable implementations of vision transformer variants
https://github.com/huggingface/pytorch-image-models (timm)	1000+ pretrained vision models; the transfer-learning workhorse
https://github.com/Lightning-AI/pytorch-lightning	Training-loop framework; read it to see what a mature loop needs
https://github.com/labmlai/annotated_deep_learning_paper_implementations	60+ papers implemented with side-by-side notes

Clone the starter set with: `bash ../_tools/clone_repos.sh 02-deep-learning`

Tools

Tool	Use
Weights & Biases / MLflow	experiment tracking — https://wandb.ai / https://mlflow.org
TensorBoard	built-in, zero setup — https://www.tensorflow.org/tensorboard
Optuna	hyperparameter search — https://optuna.org
Netron	visualise any model file — https://netron.app
Hugging Face Accelerate	device/distribution abstraction — https://huggingface.co/docs/accelerate
Google Colab / Kaggle Notebooks	free GPUs — https://colab.research.google.com

Visual explainers

- **The Illustrated Machine Learning** — <https://illustrated-machine-learning.github.io/>
- **Distill.pub** (archived but timeless) — <https://distill.pub/>
- **CNN Explainer** (interactive) — <https://poloclub.github.io/cnn-explainer/>
- **Tensorflow Playground** (build intuition in 5 minutes) — <https://playground.tensorflow.org/>